MSRF-Net: Multiscale Receptive Field Network for Building Detection From Remote Sensing Images

Yuanhao Zhao, Genyun Sun, *Senior Member, IEEE*, Li Zhang, Aizhu Zhang, *Member, IEEE*, Xiuping Jia, *Fellow, IEEE*, and Zheng Han

***Abstract*— Extracting buildings from remote sensing images plays an important role in urban development planning, disaster assessment, and mapping. Convolutional neural network (CNN) has been widely applied to building extraction because of its powerful deep semantic feature extraction ability. However, the existing CNN-based building extraction methods are difficult to accurately extract multiscale buildings with accurate edges because of the limitation of feature receptive fields and the loss of spatial detail information. For the above problems, this article proposes a multiscale receptive field network (MSRF-Net) to accurately extract multiscale buildings from remote sensing images. MSRF-Net includes multiscale receptive field feature encoder (MRFF-Encoder) and multipath decoder. In the MRFF-Encoder, a multiscale attentional down (MSAD) module and asymmetric residual inception (ARI) module are proposed to capture multiscale receptive field features. In the multipath decoder, convolutions with different kernel sizes and dilations are used in three parallel paths to learn localization-preserved multiscale features with multiscale receptive field. What is more, the features of different branches and MRFF-Encoder are fused by the proposed feature combination module, which contribute to capture context information of multiscale receptive field while recovering the resolution of feature space. The experimental results show that compared with the latest MAP-Net, MSRF-Net has achieved F1 score growth of 1.14%, 0.42%, and 1.11% and IoU score growth of 1.68%, 0.76%, and 1.64%, respectively, on Massachusetts dataset, Wuhan University (WHU) dataset, and the Typical Cities Building dataset.**

***Index Terms*— Building detection, deep learning, multiscale receptive field, semantic segmentation.**

I. Introduction

BUILDING information is of great significance to urban development, urban environmental assessment, illegal building detection, and disaster assessment [1]. With the development of sensor technology, the spatial resolution and temporal resolution of remote sensing images are continuously improved. Therefore, high-resolution image has also become one of the preferred data sources of building extraction [2], [3], [4], [5], [6]. However, increasing resolution of aerial images results in an increasing degree of redundant interference information and infernal differences. Moreover, buildings have various scales, complex materials, and diverse shapes, which make accurate extraction of buildings still a challenge [7].

Over the past two decades, building extraction has garnered significant attention from researchers. Traditional extraction methods primarily aim to extract features with high discrimination and improve classifiers [8]. For example, Jung and Schramm [9] detected the overall trend of buildings based on edge and corner detection and showed excellent results in detecting large-scale neat buildings, such as residential areas. In addition, machine learning, such as support vector machine (SVM) [10], random forest (RF) [11], and boosting [12], is widely used in building detection. To improve the detection accuracy, Zhang et al. [13] combined SAR and optical data to distinguish buildings based on SVM, and the extraction effect is obviously better than using only one kind of data. Traditional methods can efficiently and accurately extract buildings with regular shape in remote sensing images; however, they do not perform well in detection accuracy when applied to complex scenes due to the limitations of feature representation capability and classifier performance.

Recently, deep learning has become one of the most popular methods of remote sensing image processing; the commonly used methods includes deep Boltzmann machine (DBM) [14], stacked autoencoder network [15], convolutional neural network (CNN), and so on [16], [17], [18], [19], [20], [21]. Because of the powerful deep semantic feature learning ability, CNN has been widely used in object detection. For example, VGG [22] completes the extraction of depth features through continuous convolution and pooling layers. ResNet [23] and DenseNet [24] enhance model performance and accelerate model convergence by reusing semantic features. However, in order to complete the classification, CNN realizes the mapping from feature to label through the full connected layer, but the parameter redundancy of full connection makes the CNN model inefficient [25]. Evolving from CNN, semantic segmentation network replaces the fully connected layer with decoder, which can achieve end-to-end classification [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36].

Semantic segmentation is widely used in the field of building extraction because of its efficiency and feature learning capabilities [27], [28], [29], [30]. Researchers have proposed a number of methods to combine the traditional semantic

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Yuanhao Zhao, Genyun Sun, Aizhu Zhang, and Zheng Han are with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao 266580, China (e-mail: genyunsun@163.com).

Li Zhang is with the Key Laboratory of Poyang Lake Wetland and Watershed Research, Ministry of Education, Jiangxi Normal University, Nanchang 330022, China.

Xiuping Jia is with the School of Engineering and Information Technology, University of New South Wales at Canberra, Canberra, ACT 2600, Australia (e-mail: x.jia@adfa.edu.au).

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segmentation models with the scale characteristics of buildings. Ma et al. [33] proposed a local and global encoder to learn the representative features from the aerial images for describing the buildings. ENRU-Net [35] captures global contextual information through an asymmetric pyramid nonlocal block as well as self-attention mechanism and extracts it better than the traditional methods. However, deep learning loses detailed information in the original image due to downsampling during feature mining, which tends to blur the edges of the extraction results.

To alleviate the edge blur effect, most semantic segmentation networks transfer shallow features containing numerous detailed information to the decoder by skip connection [34]. For example, the efficient network (ESF-Net) [37] adopts the separable factorization and dilated convolution of the remaining blocks to maintain a slight loss of accuracy while reducing the computational cost and memory consumption. What is more, conditional random field (CRF) [28], [38], [39] was often used in the postprocessing stage, which can effectively improve the detection accuracy of edge areas. Although the extraction accuracy has been improved, postprocessing will not only increase the complexity of the process, but also the accurate extraction of the boundary in the original image is still a difficult problem. In addition, the simple structure of the decoder is difficult to make full use of the shallow features of jump connection transmission, which makes the problem of edge still a difficulty.

To accurately extract multiscale buildings, feature pyramids [36], [40], [41], which can extract features of multiscale receptive fields, are widely used for object detection and show better performance in multiscale building extraction. ARC-Net [42] introduces the residual block with asymmetric convolution (RBAC), which can reduce the computational cost for building detection. In addition, multipath network is a common method to enrich feature receptive field. HRNetv1 [43] and HRNetv2 [44] proposed a parallel network structure to solve the problem of multiscale feature extraction in model high-resolution input and achieved good performance in semantic segmentation. SIU-Net [30] designed two weight-shared branches with multiscale input to extract multiscale buildings, which improve the segmentation accuracy especially for multiscale buildings, yet obviously increased the computational complexity. However, limited by the number of pyramid layers and paths, the performance of the model is still unsatisfactory when the building scales vary too much.

Attention mechanism [45], [46], [47], [48], [49] can effectively improve the segmentation performance of deep learning model. It can capture more informative features from space or channel. He and Jiang [50] applied convolutional block attention module (CBAM) [48] to all depths in the decoder, and the building boundary information in the detection results was significantly improved. Das and Chand [51] applied CBAM and residual in each convolution layer of the encoder. Zhu et al. [52] proposed a multiple participation path neural network (MAP-Net), which applies parallel network and attention mechanism to building detection, uses various spatial pools to enhance depth features, and realizes the state-of-the-art (SOTA) results on Wuhan University (WHU) dataset [30]. Nevertheless, in previous studies, the depth of features is not considered when using attention mechanism, which make the attention mechanism not fully utilized. Therefore, the improvement of model performance by attention mechanism is limited.

For the above problems, this article proposes a multiscale receptive field network (MSRF-Net), which includes two parts: multiple receptive field feature encoder (MRFF-Encoder) and multipath decoder. MRFF-Encoder uses multiscale attentional down (MSAD) module and asymmetric residual inception (ARI) module proposed in this article to enrich the receptive fields of features. In the multipath decoder, three convolution layers with different receptive fields are used to enhance the features, and then, three independent decoders are used to restore the spatial resolution. Moreover, the decoder introduces the shallow features in the encoder through feature combination module, which helps compensate for the loss of feature details. The main contributions of this article are as follows.

1) An MSRF-Net composed of a multiscale receptive field feature encoder (MRFF-Encoder) and a multipath decoder is proposed to capture and aggregate multiscale context information, which effectively promote the buildings detection in high remote sensing images.
2) An MSAD module and an ARI module are proposed in the MRFF-Encoder to simultaneously capture global and local detailed semantic information.
3) A multipath decoder that consists of three independent decoding branches is designed to restore the spatial resolution of the features. This multipath decoder can synchronously capture the local details of features in different scales.

The remainder of this article is organized as follows. In Section II, we describe the proposed method and training strategy in detail. The experiments and results are presented in Section III. Finally, some concluding remarks are drawn in Section IV.

II. Methodology

A. Overview

For the edge and multiscale problems of buildings, in this section, we propose an MSRF-Net, which can accurately extract multiscale buildings with fine boundaries. The structure of MSRF-Net is shown in Fig. 1, and it consists of two parts.

1) MRFF-Encoder: It contains an ARI module and an MSAD module, which can capture global and local context information of features, and MSAD module contains multiscale spatial attention downsampling (MSD) and multiscale channel attention downsampling (MCD).

2) Multipath Decoder: It has three decoding paths and can combinate the features of different branches. In the following, relevant details of the above proposed network are presented.

B. Asymmetric Residual Inception (ARI) Module

It is essential to extract multiscale receptive field features for identifying buildings with large-scale differences. Over the past few years, many multiscale blocks [53] have

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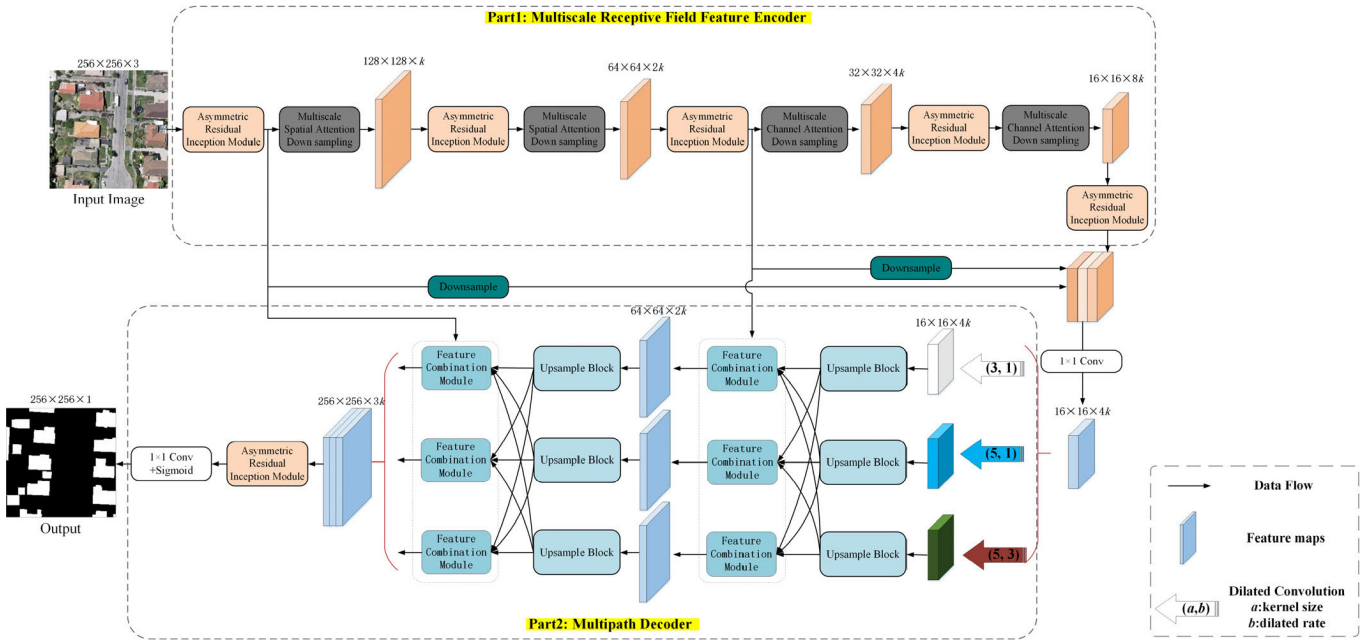


Fig. 1. Structure of the proposed MSRF-Net, which is composed of two parts: (a) multiscale receptive field feature encoder (MRFF-Encoder) and (b) multipath decoder. The parameter k is used to control the number of characteristic channels of the model. Each downsampling reduces the spatial resolution of the feature by 1/2. Each upsample block increases the spatial resolution of the feature by four times.

been proposed to extract multiscale buildings. For example, SRI-Net [54] uses the multiscale block method in the model decoding stage and has achieved good results. However, they used a large number of convolutions and did not consider the characteristics of different depth features. This makes the model parameters increase significantly and also has a negative impact on the generalization ability of the model. To resolve the above problems, we designed an efficient and high-performance ARI block, which can extract the multiscale receptive fields features.

The specific structure of ARI module is shown in Fig. 2. First, the number of channels of the input feature is adjusted from a 1×1 convolution layer and a 3×3 convolution layer to twice the original number. Then, one 1×1 convolution layers and asymmetric convolution layers with different kernels scales are used to obtain multiscale receptive field features, which is shown in the following equation:

$$\mathrm{MF} = \mathrm{Concatenate}[\mathrm{Conv}(1 \times 1), \mathrm{Concatenate}_{i=1}^{3}[\mathrm{AConv}(a_i)]]. \quad (1)$$

MF is the multiscale receptive field features, concatenate is the combination of features in channel dimension, Conv (1×1) is the features obtained by 1×1 convolution layer, and AConv(a_i) is the features obtained by asymmetric convolutional layers with kernel a_i. Asymmetric convolution decomposes the $k \times k$ convolution kernels into two convolution kernels of $1 \times k$ and $k \times 1$, which can effectively improve the efficiency of the model and reduce the computational cost. Thereafter, the residual structure is introduced into the ARI module to combine the multiscale receptive field features and the input features.

Considering the spatial scale differences of the features at different depths, three adjustable parameters a1, a2, and a3 are proposed to adjust the scale of the kernel of the asymmetric convolution layer. When the scale of the features

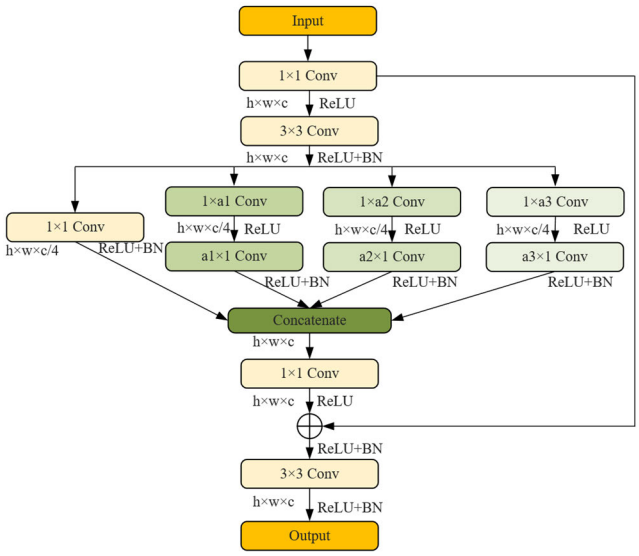


Fig. 2. Structure of asymmetric residual inception (ARI) module.

is larger, the more spatial information the features contain, therefore, the larger the convolutional kernel the better the convolution layer can perform in capturing the features, and conversely, the smaller the convolutional kernel, the better the convolution layer performs. Therefore, we set the kernel size of the asymmetric convolution to 3, 7, and 11 when the feature size is larger than 64. Otherwise, the kernel size of the asymmetric convolution is set to 3, 5, and 7.

Compared with the previous methods, ARI module employs asymmetric convolution and reduces the number of channels in multiscale receptive field features, effectively addressing the problem of parameter redundancy. In addition, ARI module uses larger asymmetric convolution with convolution kernel for large-scale features and smaller asymmetric convolution with

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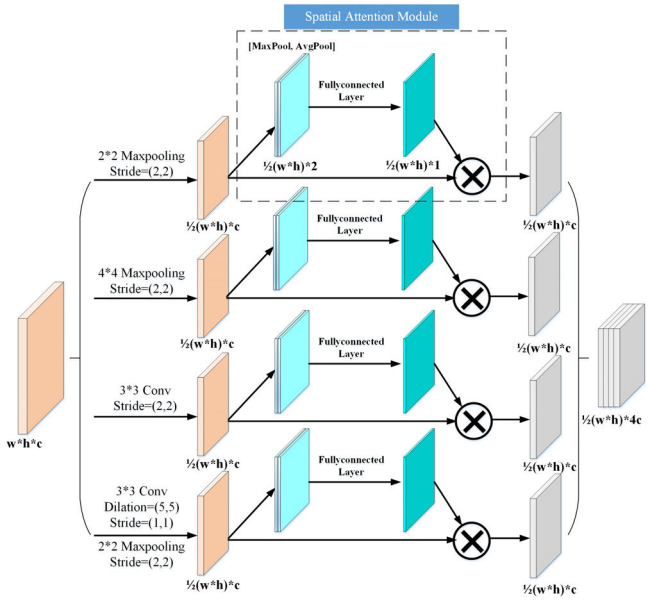


Fig. 3. Structure of multiscale spatial attention downsampling (MSD). First, four downsampling features with different receptive fields are obtained through different downsampling layers. Then, the spatial information of each group of features is enhanced based on the spatial attention mechanism (SAM). Finally, the features are stacked.

convolution kernel for small-scale features, which can capture the global and local spatial relations for more precise edge detection and better overall integrity of extracted buildings.

C. Multiscale Attentional Down (MSAD) Module

The receptive field of features obtained by traditional downsample layers is fixed, which makes it difficult for the model to capture the spatial information of multiscale buildings. A few studies have combined multiscale methods to obtain features [45], [46], [47], [48], [49], [50], [51], [52]. However, they do not consider the spatial information and channel information contained in the features at different scales. In this section, we design an MSAD module to acquire multiscale receptive field downsampled features based on attention mechanism. The MSAD module consists of two parts, MSD and MCD. MSD captures the multiscale localization detail information of features based on spatial attention mechanism (SAM), which helps to identify multiscale buildings. MCD can make the best combination of multiscale receptive field features based on the channel attention mechanism (CAM), which can make full use of the building global semantic information of different receptive field features.

The structure of MSD is shown in Fig. 3. It consists of four lower sampling layers, two maxpooling layers with kernel sizes of 2 and 4, a 3×3 convolution layer, and a 3×3 atrous convolution layer with a dilation ratio of 5. The step size of the four layers is 2. Compared with the input features, the receptive fields of the four groups of downsampled features are 2, 3, 4, and 11, respectively, which helps the model to obtain the spatial information of different buildings and make the extraction results have better integrity. The SAM is an adaptive spatial region selection mechanism that assigns a weight value to each pixel by solving for the average and maximum of all channel feature values of the pixel, which allows the model to focus on the local spatial details of the feature [48]. Therefore, MSD captures the spatial local information in four groups of downsampled features through the spatial attention module. Finally, multireceptive field features are stacked.

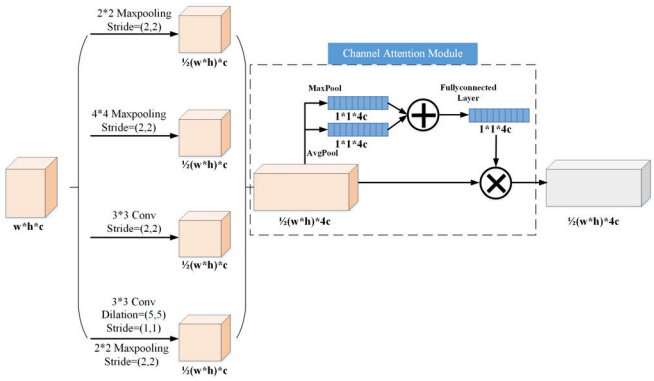


Fig. 4. Structure of multiscale channel attention downsampling (MCD). First, four downsampling features with different receptive fields are obtained through different downsampling layers. Then, the features are stacked. Finally, the optimal feature is reconstructed based on the channel attention mechanism (CAM).

The structure of MCD is shown in Fig. 4. It contains four groups of lower sampling layers, the same as MAD. The difference is that MCD uses CAM to replace the SAM. The CAM measures the amount of information in each channel of a feature by calculating the mean or maximum value [48]. Thus, CAM can adaptively recalibrate the weight of each channel, collect global information, capture channelwise relationships, and improve representation ability. Therefore, MCD directly stacks multiscale receptive field features, and then, channelwise combination is performed based on CAM.

The features of different scales contain different spatial information and feature channels. For example, the shallow features have abundant detail information and high spatial resolution. The deep features have low spatial resolution and generous feature channels. Hence, in this article, when the input feature space resolution is bigger than 64, MSD is used to downsample the features, which make the model focus on the building area and reduce the impact of noise; conversely, MCD is used, which is helpful to obtain global information about the features of multiscale receptive fields. In general, MSD and MCD can not only enrich the receptive field of features but also capture the spatial semantic information of features of different scales based on attention mechanism and combine multiscale receptive field features.

D. Multipath Decoder

Decoders is an integral part of the semantic segmentation, which can recover feature spatial resolution. However, the decoders in previous studies have simple structures and are limited by the receptive field [36], [40], [41], [42], [43], [44], which make it difficult to extract multiscale buildings accurately. Therefore, we design a multipath decoder, aiming to capture more contextual information and compensates for the loss of detail information of deep semantic features by combining features with multiscale receptive fields.

The specific structure of the multipath decoder is shown in Fig. 5. The gray features in Fig. 5 refer to the multiscale

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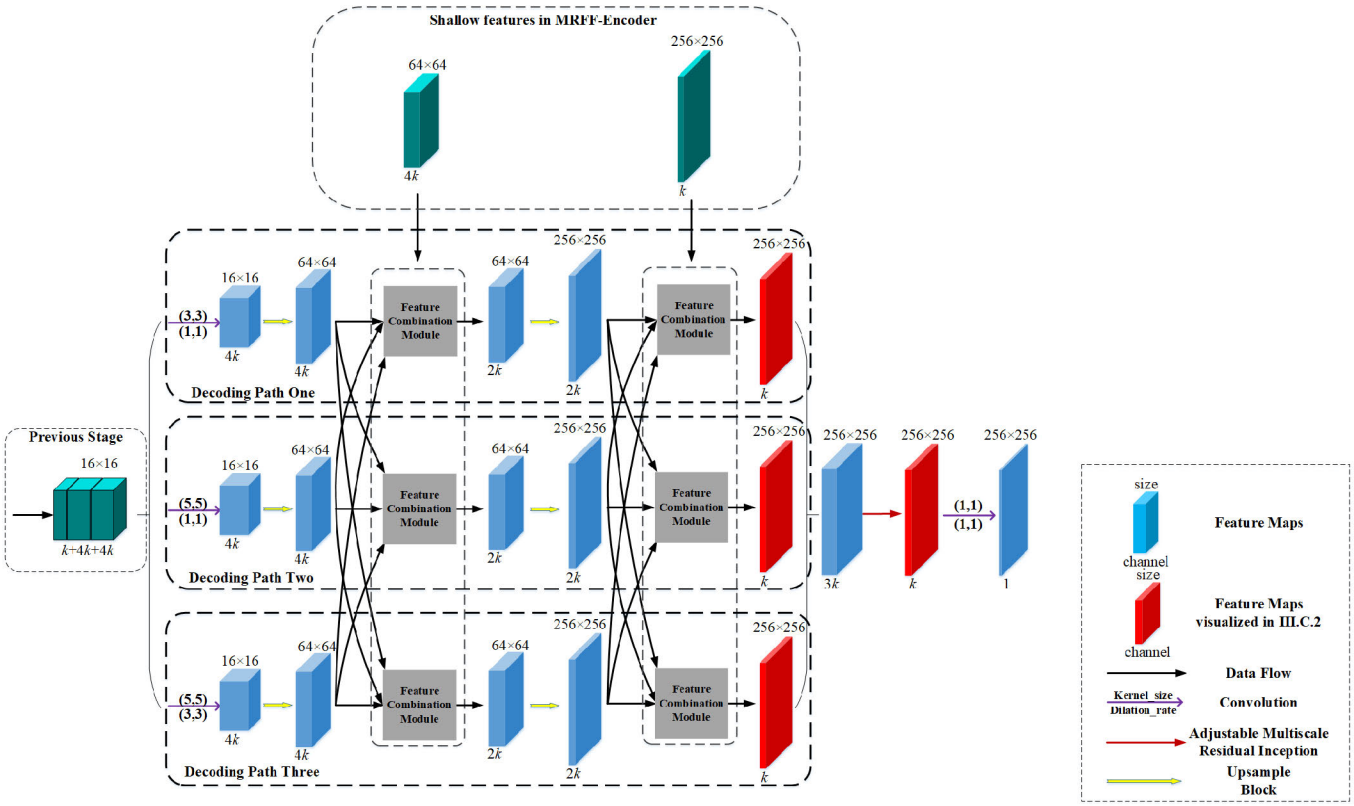


Fig. 5. Structure of multipath decoder. There are three independent decoders in the multipath decoder. Each encoder contains two upsample blocks, and each upsample block can increase the feature space resolution by 4 times. What is more, the feature combination module can combine different branches features and the features in encoder to further expand the receptive field of each branch feature.

features in MRFF-Encoder, and their introduction can effectively mitigate the loss of feature detail information. The red features in Fig. 5 are visualized in Section III-C2 to demonstrate the effectiveness of the multipath decoder. Multipath decoder contains three independent decoding branches, each with different receptive fields for the input features. Each decoding path contains two upsample blocks, and the spatial resolutions of input features and output features of each decoding path are 16 × 16 and 256 × 256, respectively. To expand the receptive field of each branch feature, we propose a feature combination module, which can combine features from other paths and encoding processes.

The structure of the upsample block and feature combination module is shown in Fig. 6. Each upsample block consists of one upsample layer and two 3 × 3 convolution layers, and the upsample method used in this article is bilinear upsample, which has few parameters and fast calculation speed. The feature combination module reduces the channels of other features by half through 1 × 1, which not only compensates for the loss of detail information caused by downsampling but also reduces the noise impact of shallow features and further extends the receptive field of features of different paths. Finally, we combine multiscale receptive field features by skip concatenate, which is a feature combination method that stacks features at different locations in the channel dimension. Skip concatenate can better preserve the semantic information of different receptive field features.

Due to the large difference in the receptive field of each path input feature in the multipath decoder, different paths

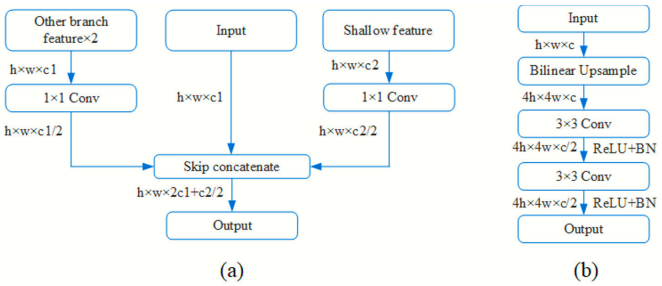


Fig. 6. Structure of (a) feature combination module and (b) upsample block.

show different effects in extracting buildings at different scales, which are compared in Section III-C2. In addition, the introduction of shallow features and the combination of different path features make the buildings in the extraction results have more accurate edge information.

E. Training Strategy

To improve the generalization and prediction ability of the model, the activation function of the convolution layer in the model is set as RELU function. The output layer of MSRF-Net is a single-channel feature map with the same spatial resolution as the input image, and each value represents the probability that it belongs to the building.

All the researches were implemented on TensorFlow using a single 2080 GPU with 8-GB memory. Adam is selected as the optimizer. The initial learning rate is 0.001, and if the model validation accuracy is not improved within four epochs, the learning rate will be reduced by half. All the

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comparison methods are trained for 50 epochs. What is more, the batch size is set to 8. In the comparative experiment of this article, all methods use the same superparameters to ensure the effectiveness of the comparison results.

Building extraction based on semantic segmentation can be regarded as pixelwise binary classification, so this article selects binary cross entropy as the loss function of the model. The calculation formula of loss function is shown in the following equation:

$$\text{loss} = -\frac{1}{N}\sum_{i=1}^{N}\left(y_i \log(p(y_i)) + (1-y_i)\log(1-p(y_i))\right) \tag{2}$$

where N is the number of pixels contained in the image, and the loss of the whole image is the average value of all pixel loss values. y is the value in the label image, which is the value of 0 and 1; $p(y)$ is the value in the prediction result, and $p(y) \in [0,1]$.

III. Experiment and Analysis

A. Dataset

We evaluated the performance of the proposed method on three open datasets, including the WHU building dataset [30], the Massachusetts dataset [55], and the dataset of building instances of typical cities in China [56]. The details of these datasets are described as follows.

The WHU building dataset [30] includes aerial and satellite remote sensing images with the corresponding label images. In this experiment, we choose aerial image datasets, which contains more diverse buildings and training samples to evaluate the robustness of the proposed algorithm. It consists of more than 187 000 buildings, covering over a 450 km^2 area with 0.3-m spatial resolution. Each image has three bands, corresponding to red (R), green (G), and blue (B) wavelengths, with each image size of 512 × 512 pixels. There is a total of 8188 tiles of images, including 4736, 2416, and 1036 tiles as training, test, and validation datasets, respectively. We conducted our experiment at its original provided dataset partitioning. In order to obtain more data and fully complete the training of the model, we cut each remote sensing image into four tiles with the size of 256 × 256 pixels.

The Massachusetts building dataset [55] consists of 151 aerial images of the Boston area. The size of each image in dataset is 1500 × 1500 pixels, and it covers approximately 340 km^2 with a resolution of 1 m. It consists of 137 sets of aerial images and corresponding single-channel label images for training part, 10 for testing part, and 4 for validation part. We clipped the images of training set, validate set, and test set to 256 × 256 pixels, with a clipping step of 256.

The dataset of building instances of typical cities in China (Typical Cities Building dataset) [56] was published in 2021, and the samples in the dataset were distributed in Beijing, Shanghai, Shenzhen, and Wuhan. The original data are 19 satellite images provided by Google, with a ground resolution of 0.29 m, and the sample coverage area of the dataset is 120 km^2. There is a total of 7260 tiles of images, including 5985 and 1275 tiles as training and test datasets, respectively. Each satellite image is cut into tiles with the size of 500 × 500 pixels. In this experiment, we used 1/5 of the training sample as the validation dataset and cut each remote sensing image into four tiles with the size of 256 × 256 pixels.

B. Evaluation Metric

Generally, evaluation metric methodologies can be divided into two categories: pixel-level metrics and instance-level metrics. Since this article adopts the end-to-end classification method, which can classify each pixel of the input image as building or nonbuilding, we use the pixel-level metrics for performance evaluation. The evaluation metrics include overall accuracy (OA), precision, recall, F1-score (F1), and intersection over union (IoU).

In the pixel-level classification, the classification of each pixel can be summarized into four classifying conditions: true prediction on a positive sample (TP), false prediction on a positive sample (FP), true prediction on a negative sample (TN), and false prediction on a negative sample (FN). What is more, precision represents the percentage of TP in the pixels predicted as buildings, recall represents the percentage of TP in the pixels as buildings in the label, the F1 is the weighted average of precision and recall, and IoU is the average value of the intersection of the prediction and ground truth over their union of the whole image set. Equations are given as follows:

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{FP}+\text{TN}+\text{FN}} \tag{3}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{4}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{5}$$

$$\text{F1} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision+Recall}} \tag{6}$$

$$\text{IoU} = \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall} - \text{Precision} * \text{Recall}}. \tag{7}$$

Floating point operations (FLOPs) and Para (trainable model parameters) are used in this article to measure the efficiency and complexity of a model's operation, respectively. FLOPs are the number of calculations per second the computer performs while the model is running and are often used to estimate execution performance in scientific computing fields that use a large number of FLOPs. Para is the number of parameters that deep learning uses during training to adjust values based on gradients.

C. Parameter Analysis

1) Network Parameter: The parameter k is introduced to control the number of feature channels. Without losing generality, we designed an experiment to explore the potential impact of k on the performance and complexity of MSRF-Net on the Typical Cities Building dataset. The OA and IoU were used to evaluate the performance of model, and the trainable parameters (Para) and floating-point operations per second (FLOPs) were counted to represent the complexity of model. Empirically, the parameter k was set from 28 to 40 in our experiments that consider the tradeoff between accuracy and complexity.

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TABLE I

PERFORMANCE, COMPLEXITY, AND EFFICIENCY OF MSRF-NET WITH DIFFERENT K VALUES ON THE TYPICAL CITIES BUILDING DATASET

k	OA(%)	IoU(%)	Para (million)	FLOPs (Billion)
28	93.64	71.92	**6.86**	**44.61**
30	93.64	71.98	7.87	51.06
32	93.75	72.37	8.96	58.20
34	93.69	72.32	10.11	65.58
36	93.86	72.41	11.33	73.70
38	**93.89**	**72.86**	12.62	81.92
40	93.77	72.44	13.99	90.96

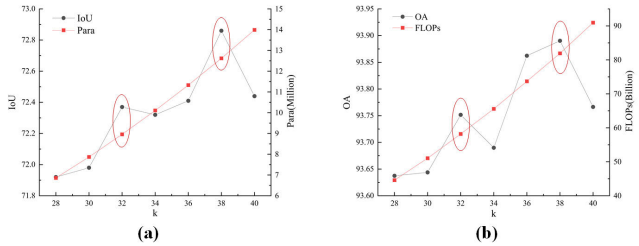


Fig. 7. (a) IoU and Para and (b) OA and FLOPs of MSRF-Net with different k values on the Typical Cities Building dataset.

The specific experimental results are shown in Table I and Fig. 7. The increase of k means that features will contain more feature channels and have stronger semantic information extraction capability. However, having too many feature channels can also negatively impact the model's generalization capability, resulting in reduced performance. Therefore, the performance of the model shows an approximate trend of first increasing and then decreasing as the parameter k increases. Moreover, increasing k and the number of feature channels will increase the complexity of the model and causing the model parameters to increase as well.

When k equals 32 and 38, the model performance achieves a local optimal result, as shown in the red oval area in Fig. 7. When k is equal to 32, the Para is less than 10M, and the accuracy of the model is better than that of the models when k equals 34 and 36. When k is equal to 38, although the Para and FLOPs have increased significantly, the amount of model parameters is still less than that of other comparison methods, and the performance of the model has also been significantly improved. Therefore, in order to better balance the performance and complexity of the model, we set k to 32 and 38, respectively, in the subsequent experiments.

2) Parameters of ARI Module: The ARI module in MSRF-Net can learn multiscale receptive field features based on convolutional layers with different kernel scales. a1, a2, and a3 are the kernel scales of the convolutional layers in the ARI module. Different parameters of a1, a2, and a3 affect the classification effect of the model. In order to select the optimal parameters, we set up five sets of comparison experiments with different scales of convolutional layers, and the parameters settings for each group comparison tests are shown in Table II. In Model-S and Model-L, the parameters a1, a2, and a3 are fixed and do not change based on the feature scale; in the other three sets of experiments, the parameters a1, a2, and a3 vary based on feature scale.

The specific experimental results are shown in Table III and Fig. 8. Model-S uses a smaller convolution kernel, and the model has higher classification accuracy, but there may be problems with some large buildings that are difficult to identify. Model-L shows better classification results for large buildings, as shown in Fig. 8(c) and (d). The classification accuracy of Model-AS and Model-AL is low, and the extraction effect of multiscale buildings and the retention effect of building edges are both poor. The specific results are shown in Fig. 8(e) and (f). This may be due to the fact that too large or too small convolution kernels cannot accurately capture the spatial features of multiscale buildings in the image when learning features. Compared to the classification results of the four comparison models, MSRF-Net can accurately identify multiscale buildings in remote sensing images, so the parameter settings in the ARI module are reasonable and effective.

3) Significance of Multipath Decoder: In order to verify the significance of multipath decoder, we visualized the feature maps of different decoding paths and the feature maps combined three decoding path feature maps, and the visualized features are represented by red cubes in Fig. 5. The feature maps in multipath decoder contain abundant feature channels. Therefore, to visualize features more intuitively, we reduce the dimension of the feature map to include only one feature channel through PCA transformation.

Fig. 9 shows the feature maps extracted from the proposed multipath decoder with different paths. The size of all feature maps is 256 × 256. Columns (c)–(e) show the feature maps extracted from the different decoding paths. The feature maps extracted from different decoding paths have different receptive field and show different results of multiscale buildings. Feature maps with smaller receptive fields perform better in extracting small buildings but can lead to hollowness in large buildings. Conversely, feature maps with larger receptive fields can extract large buildings well but may suffer from the loss of detail information and building edge blurring. Column (f) shows the feature map, which combine three decoding path feature maps. Multipath decoder combines multiscale receptive field features in different decoding paths, which can capture more abundant semantic information and perform better when extracting multiscale buildings. Therefore, the features obtained by the proposed multipath decoder have better effect in detail preservation, as shown in the red rectangular area. In addition, the combined features can accurately maintain the edges of multiscale buildings, as shown in the red oval area.

4) Performance Evaluation: To evaluate the performance of the proposed network, we conducted contrast experiments to compare MSRF-Net with the most recent building detection methods, including FCN [32], SegNet [26], U-Net [34], PSPNet [36], HRNetv2 [44], EU-Net [40], and MAP-Net [52] on WHU building dataset [30], Massachusetts building dataset [55], and Typical Cities Building dataset [56].

FCN is the first proposed semantic segmentation network, which uses convolution layer to replace full connection layer to realize end-to-end classification. SegNet and U-Net combine

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TABLE II
SETTINGS OF PARAMETERS A1, A2, AND A3 IN THE FIVE GROUPS OF COMPARISON EXPERIMENTS

	Model-S	Model-L	Model-AS	Model-AL	MRSF-Net
Feature scale>64	3,5,7	3,7,11	3,5,7	5,11,17	3,7,11
Feature scale<= 64	3,5,7	3,7,11	1,3,5	3,7,11	3,5,7

TABLE III
COMPARE OF MODEL ACCURACY WITH DIFFERENT ARI MODULE PARAMETERS ON THE MASSACHUSETTS DATASET

Model	OA(%)	Precision(%)	Recall(%)	F1(%)	IoU(%)
Model-S	94.01	86.53	80.37	83.34	71.43
Model-L	93.96	85.78	**81.00**	83.32	71.41
Model-AS	93.87	**87.21**	78.65	82.71	70.52
Mdoel-AL	93.40	84.05	79.74	81.84	69.26
MSRF-Net	**94.09**	86.53	80.88	**83.61**	**71.84**

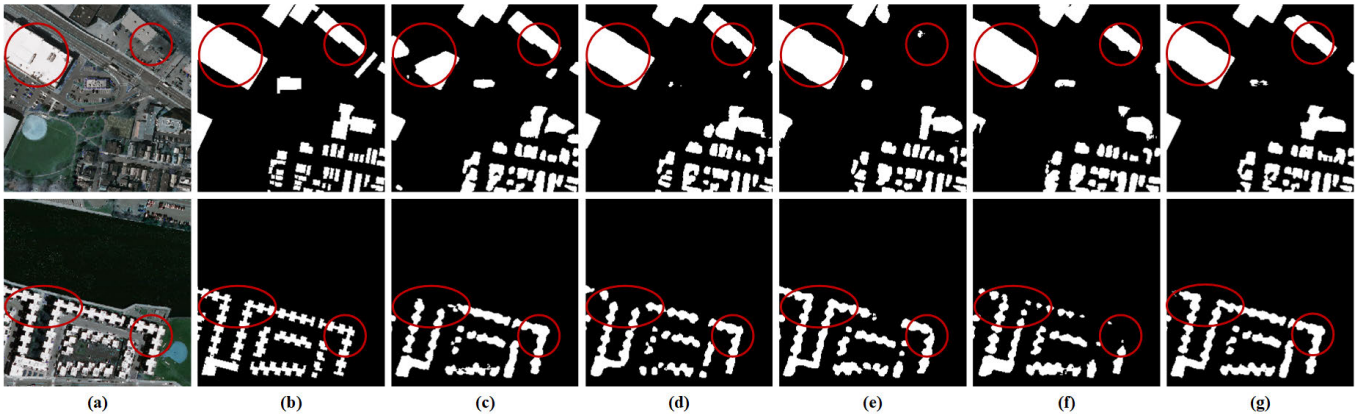


Fig. 8. Example of building extraction results produced by different methods on the Massachusetts dataset. (a) Original image. (b) Label image. (c) Model-S. (d) Model-L. (e) Model-AS. (f) Model-AL. (g) MSRF-Net.

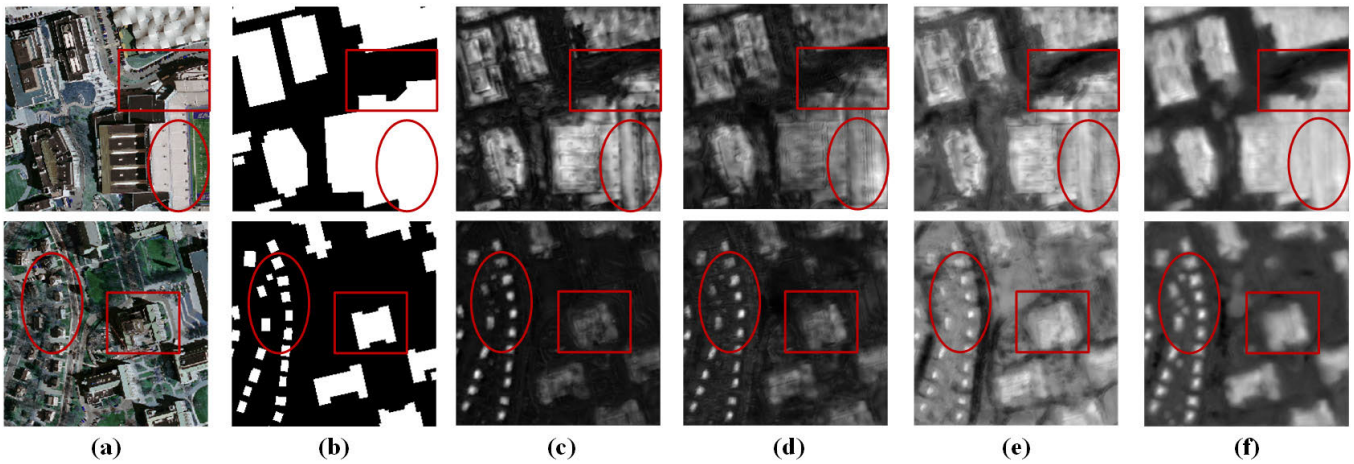


Fig. 9. Feature maps extracted from multipath decoder with different paths. (a) Original image. (b) Label image. Feature maps extracted from (c) decoding path one, (d) decoding path two, and (e) decoding path three. (f) Feature map combined three decoding path feature maps.

the features of different depths by skip connections and pooling indices, which effectively reduce the loss of detail of deep features. PSPNet proposes a pyramid pooling module, which can aggregate the context information of different regions. EU-Net designs the dense spatial pyramid pooling (DSPP) to extract dense and multiscale buildings. HRNetv2 and MAP-Net are recently proposed multipath networks. HRNetv2 combines the features of different depths through four branches to complete the fusion of multiscale features. MAP-Net uses three independent paths to combine different scale features in the encoding part.

Experimental results are listed in Tables IV–VI, and the best records are marked with bold. It can be seen that our method achieved IoU of 72.27%, 90.16%, and 72.86% on

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TABLE IV
COMPARISON OF THE STATE-OF-THE-ART METHODS AND OURS ON THE MASSACHUSETTS DATASET

Model	OA(%)	Precision(%)	Recall(%)	F1(%)	IoU(%)
FCN	92.21	81.99	74.58	78.11	64.08
SegNet	91.54	79.41	73.74	76.47	61.90
U-Net	90.35	69.54	**85.76**	76.80	62.34
PSPNet	92.43	83.85	73.57	78.37	64.44
HRNetv2	93.39	83.95	79.79	81.82	69.23
EU-Net	93.44	84.45	79.41	81.85	69.28
MAP-Net	93.80	85.94	79.80	82.76	70.59
MSRF-Net(k=32)	94.09	86.53	80.88	83.61	71.84
MSRF-Net(k=38)	**94.21**	**87.04**	80.98	**83.90**	**72.27**

TABLE V
COMPARISON OF THE STATE-OF-THE-ART METHODS AND OURS ON THE WHU DATASET

Model	OA(%)	Precision(%)	Recall(%)	F1(%)	IoU(%)
FCN	98.34	92.29	92.84	92.56	86.16
SegNet	98.36	93.42	91.71	92.56	86.14
U-Net	98.40	94.50	90.88	92.66	86.32
PSPNet	98.59	93.19	94.21	93.70	88.14
HRNetv2	98.66	93.40	94.59	94.00	88.67
EU-Net	98.69	93.76	94.54	94.15	88.94
MAP-Net	98.75	93.99	94.82	94.40	89.40
MSRF-Net(k=32)	98.83	94.50	**95.00**	94.75	90.02
MSRF-Net(k=38)	**98.85**	**94.97**	94.68	**94.82**	**90.16**

TABLE VI
COMPARISON OF THE STATE-OF-THE-ART METHODS AND OURS ON THE TYPICAL CITIES BUILDING DATASET

Model	OA(%)	Precision(%)	Recall(%)	F1(%)	IoU(%)
FCN	91.68	80.36	76.02	78.13	64.11
SegNet	92.28	81.10	78.89	79.98	66.64
U-Net	91.28	76.93	79.13	78.01	63.95
PSPNet	93.17	85.49	78.34	81.76	69.15
HRNetv2	93.56	84.87	81.58	83.19	71.22
EU-Net	92.98	83.14	80.36	81.73	69.10
MAP-Net	93.66	**86.73**	79.80	83.12	71.11
MSRF-Net(k=32)	93.75	84.20	83.74	83.97	72.37
MSRF-Net(k=38)	**93.89**	84.67	**83.94**	**84.30**	**72.86**

three datasets, which is 1.68%, 0.76%, and 1.75% higher than the latest MAP-Net. In addition, the OA, F1, and IoU of our method are all optimal values in all comparison methods of the three datasets.

To compare the extraction results of different methods, the example results of five models with better accuracy on three datasets are given in Figs. 10–12. Columns (a) and (g) represent the original image and corresponding ground truth, and columns (b)–(f) are the extracted results from PSPNet, HRNetv2, EU-Net, MAP-Net, and MSRF-Net, respectively.

Overall, the better performance of the proposed MSRF-Net is clearly perceivable. MSRF-Net can not only accurately

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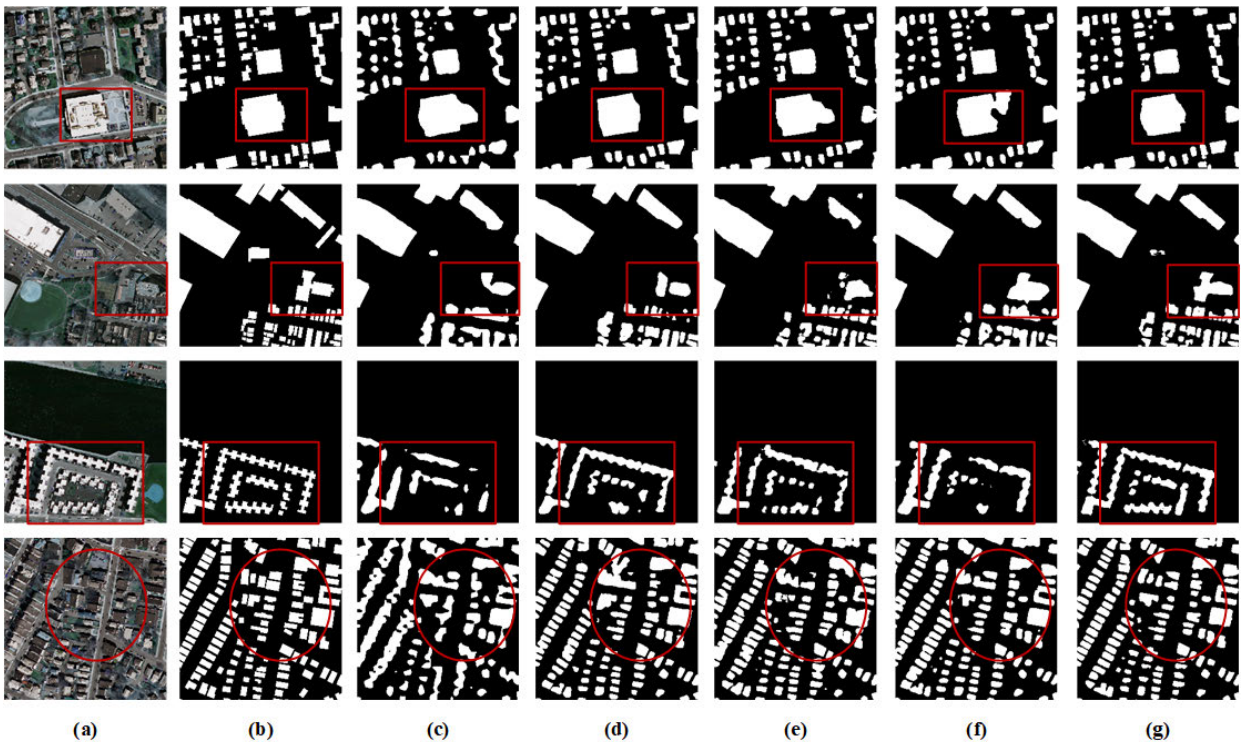


Fig. 10. Example of building extraction results produced by different methods on the Massachusetts dataset. (a) Original image. (b) Ground truth. (c) PSPNet. (d) HRNetv2. (e) EU-Net. (f) MAP-Net. (g) MSRF-Net ($k = 32$).

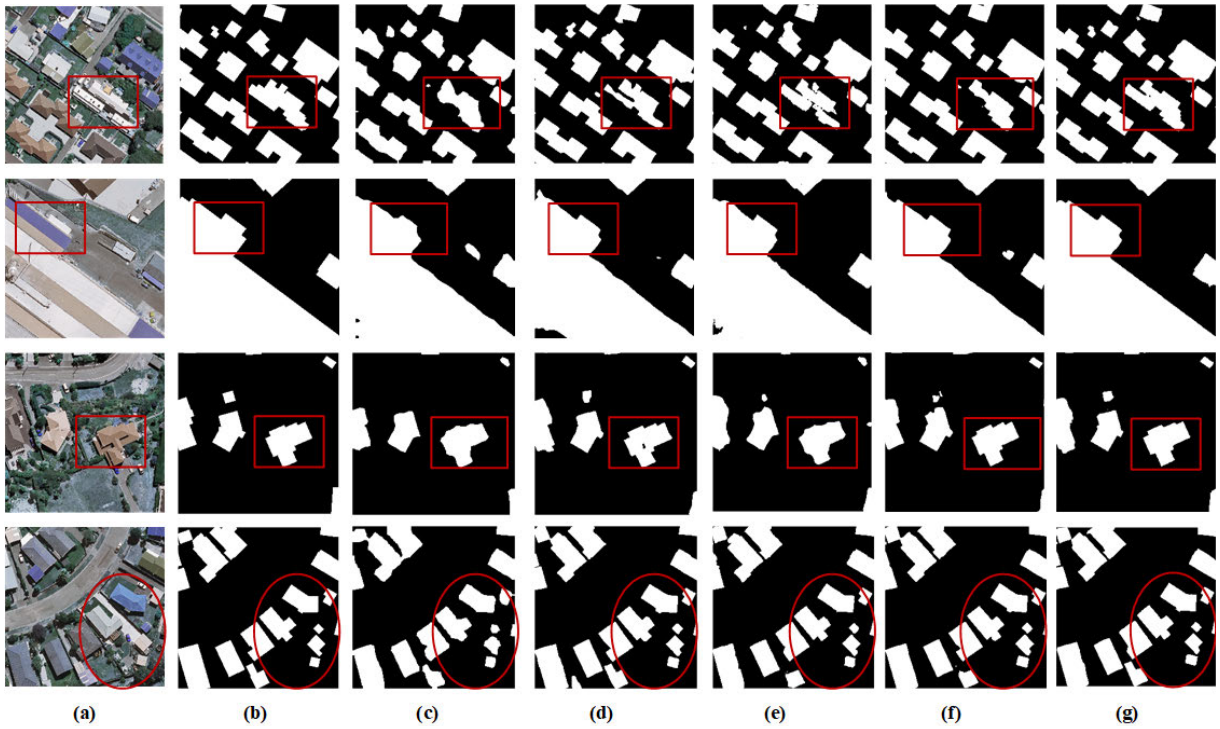


Fig. 11. Example of building extraction results produced by different methods on the WHU dataset. (a) Original image. (b) Ground truth. (c) PSPNet. (d) HRNetv2. (e) EU-Net. (f) MAP-Net. (g) MSRF-Net ($k = 32$).

extract multiscale buildings but also more accurately maintain the edges of buildings. In addition, MSRF-Net effectively controls the problem of wrong and missing classifications. Specifically, as shown in the red rectangle area, MSRF-Net has excellent performance in edge area for multiscale buildings and can distinguish adjacent small buildings. In addition, MSRF-Net can accurately identify buildings with large-scale difference in remote sensing images, as shown in the red oval area. What is more, MSRF-Net can accurately identify the small buildings surrounded by a large number of vegetation and will not identify the impermeable surface with regular shape as a building, while other methods cannot solve the above problems, as shown in the yellow oval area and yellow rectangular area.

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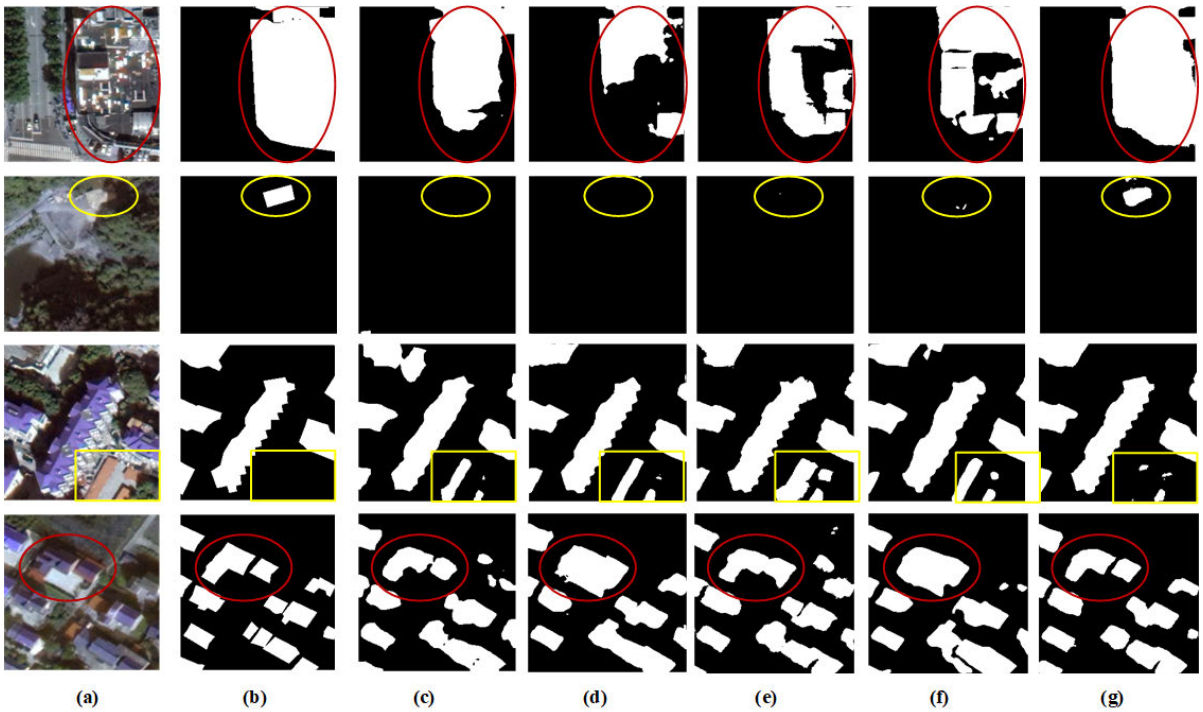


Fig. 12. Example of building extraction results produced by different methods on the Typical Cities Building dataset. (a) Original image. (b) Ground truth. (c) PSPNet. (d) HRNetv2. (e) EU-Net. (f) MAP-Net. (g) MSRF-Net ($k = 32$).

TABLE VII

INFLUENCE OF DIFFERENT MODULES ON MASSACHUSETTS DATASET. MSRF-NET/MSAD: MSRF-NET WITHOUT MSAD MODULE. MSRF-NET/ARI: MSRF-NET WITHOUT ARI MODULE. MSRF-NET/MULTIPATH DECODER: MSRF-NET WITHOUT MULTIPATH DECODER

Model	OA(%)	Precision(%)	Recall(%)	F1(%)	IoU(%)	Para (Million)
MAP-Net	93.80	85.94	79.80	82.76	70.59	24.00
MSRF-Net/MSAD	93.87	8.75	80.48	83.03	70.98	5.84
MSRF-Net/ARI	94.00	86.58	80.21	83.28	71.34	9.11
MSRF-Net/Multipath Decoder	93.90	86.01	80.31	83.07	71.04	7.33
MSRF-Net (k=32)	94.09	86.53	80.88	83.61	71.84	8.96

5) Ablation Experiments: To explore the contributions of MSAD module, ARI module, and multipath decoder on the MSRF-Net, we perform ablation experiments by removing and changing each critical component of the network independently on the Massachusetts dataset and evaluate the accuracy using OA, precision, recall, F1, IoU, and Para.

To evaluate the influence of MSAD Module, we replace the MSAD module with a maxpooling layer with step size 2, which is called MSRF-Net/MSAD. Then, we replace the ARI module with two consecutive 3×3 convolution layers, which is called MSRF-Net/ARI, which is used to evaluate the influence of ARI module. Finally, we explore the effect of multipath decoder. We remove the multipath decoder in MSRF-Net and combine multiscale receptive field features with a feature pyramid structure. The decoder structure is the same as that of each independent decoder in the multipath decoder. This model is called MSRF-Net/multipath decoder. The parameter k in all the ablation experiment was set to 32.

Experimental results are shown in Table VII and Fig. 13. Note that all settings and metadata for training and validation processes are fixed in all ablation experiments. Both MASD and ARI modules expand the receptive field of features in the encoding part by extracting multiscale features; therefore, adding MASD and ARI modules improved the IoU by 0.86% and 0.5%, respectively, and also effectively improves the model's extraction of multiscale buildings, as shown in the red oval area in Fig. 13. The multipath decoder recovers the spatial resolution of features through three decoders and combines the features in different decoders and the shallow features in the MRFF-Encoder. According to the experimental results, adding the multipath decoder improved the F1 and IoU by 0.54% and 0.8%, respectively. It can not only make the model better maintain the edge details of buildings but also effectively distinguish adjacent buildings, as shown in the red rectangular area in Fig. 13.

The MSAD module and the multipath decoder use multiple parallel paths and more downsampling layers to obtain multiscale receptive field features; therefore, the use of these two modules increases the model parameters along with the accuracy. The ARI module reduces the number of channels in the multiscale receptive field because it uses 1×1 convolution;

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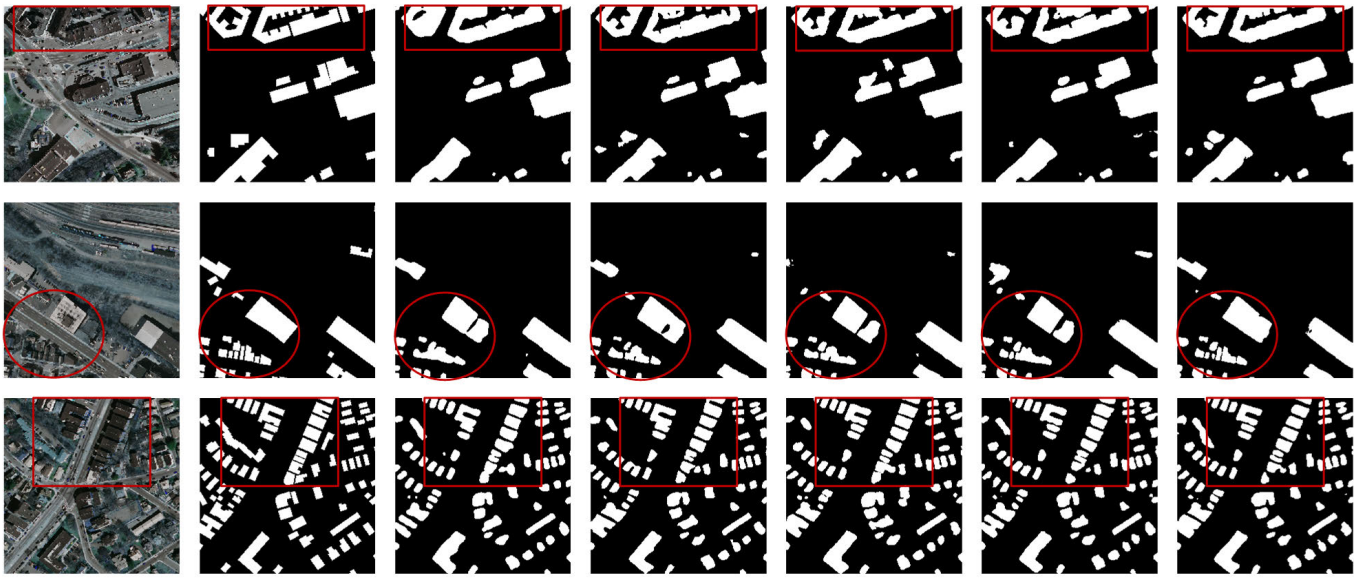


Fig. 13. Example of building extraction results on ablation experiments. (a) Original image. (b) Ground truth. (c) MAP-Net. (d) MSRF-Net—ARI. (e) MSRF-Net—MSAD. (f) MSRF-Net—multipath decoder. (g) MSRF-Net ($k = 32$).

TABLE VIII
COMPARISON OF RELATED METHODS ON F1, IOU, PARA, AND FLOPS ON THE TYPICAL CITIES BUILDING DATASET

Model	F1(%)	IoU(%)	Para (Million)	FLOPs (Billion)
FCN	78.13	64.11	14.72	**40.12**
Segnet	79.98	66.64	31.82	106.59
U_Net	78.01	63.95	31.06	109.41
PSPNet	81.76	69.15	48.30	738.28
HRNetv2	83.19	71.22	30.13	175.35
EU-Net	81.73	69.10	26.85	45.52
MAP-Net	83.12	71.11	25.57	374.71
MSRF-Net(k=32)	83.97	72.37	**8.96**	58.20
MSRF-Net(k=38)	**84.30**	**72.86**	12.62	81.92

therefore, the use of ARI module increases the model accuracy and reduces the model parameters.

6) Complexity of MSRF-Net: To validate the tradeoff between the performance and complexity of MSRF-Net, we compared F1, Para, IoU, and FLOPs of related methods on the Typical Cities Building dataset.

The quantitative results are shown in Table VIII, from which we can find that FCN has the lowest complexity in the previous methods but with poor performance. PSPNet has the most Para and FLOPs in the previous methods, because the encoder of PSPNet is based on ResNet, which contains a large number of convolution layers. Some paths in MAP-Net needed to process feature maps with large resolution to preserve exact localization in the whole network, which lead to large numbers of Para. EU-Net contains fewer convolution layers and more feature channels; therefore, although its Para is more than MAP-Net, its FLOPs is obviously less than MAP-Net.

Compared with other methods, MSRF-Net uses fewer convolution layers to extract the features of multiscale receptive fields. Although the multipath decoder has three branches, each decoder contains only five convolution layers. Therefore, MSRF-Net contains less parameters. As shown in Table VIII, the FLOPs of MAP-Net are only more than FCN and EU-Net. Compared with MAP-Net, which has the second highest accuracy, the Para of MSRF-Net ($k = 32$) is 1/3 of MAP-Net and the FLOPs of MSRF-Net are 1/6 of MAP-Net. Compared with other related methods, MSRF-Net can maintain higher accuracy and lower complexity.

IV. CONCLUSION

In this article, we propose an MSRF-Net to extract multiscale buildings. In the MSRF-Net, MSAD module and ARI modules are proposed to enrich the receptive field of features. In addition, a multipath decoder is designed to restore the spatial resolution of feature and capture multiscale context information, which is helpful to extract multiscale buildings. The experimental results on three datasets show that the proposed method outperforms a number of SOTA approaches, including HRNetv2, EU-Net, and MAP-Net.

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Although MSRF-Net has achieved better results in building extraction, it still has some limitations: 1) MSRF-Net uses complete semantic segmentation tags, and the production of tags requires a lot of human input, which makes MRSF-Net require high annotation collection costs when applied; 2) MSRF-Net has more parameters, and the determination of the parameters requires more experiments; and 3) the recognition effect for single small buildings in mountainous areas or forests still needs to be improved. Based on the above problems, future work in this study will focus on extending the framework to semi-supervised or unsupervised frameworks, parameter adaption, and detailed information mining.

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Yuanhao Zhao received the B.Sc. degree in geomatics engineering from the China University of Petroleum (East China), Qingdao, China, in 2020, where he is currently pursuing the Ph.D. degree in marine resources and information engineering.

His research interests include feature extraction and remote sensing applications in cities.



Genyun Sun (Senior Member, IEEE) received the B.Sc. degree from Wuhan University, Wuhan, China, in 2003, and the Ph.D. degree from the Institute of Remote Sensing Applications, Chinese Academy of Sciences, Beijing, China, in 2008.

He is currently a Professor with the China University of Petroleum (East China), Qingdao, China. His research interests cover remote sensing image processing, including hyperspectral and high-resolution remote sensing, and intelligent optimization.



Li Zhang received the B.Sc., M.Sc., and Ph.D. degrees from Wuhan University, Wuhan, China, in 2003, 2008, and 2016, respectively.

She is currently an Associate Professor with Jiangxi Normal University, Nanchang, China. Her research interests cover remote sensing application and remote sensing image processing, including wetland remote sensing, hyperspectral, and high-resolution remote sensing.



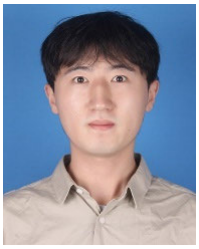
Aizhu Zhang (Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees from the China University of Petroleum (East China), Qingdao, China, in 2011, 2014, and 2017, respectively.

She is currently a Lecturer with the China University of Petroleum (East China), Qingdao, China. Her research interests are in artificial intelligence, pattern recognition, city remote sensing, and wetland remote sensing.



Xiuping Jia (Fellow, IEEE) received the B.Eng. degree from the Beijing University of Posts and Telecommunications, Beijing, China, in 1982, and the Ph.D. degree in electrical engineering from the University of New South Wales, Kensington, NSW, Australia, in 1996.

Since 1988, she has been with the School of Engineering and Information Technology, The University of New South Wales at Canberra, Campbell, ACT, Australia, where she is currently an Associate Professor. She has authored or coauthored more than 270 referred articles, including over 160 journal articles, addressing various topics, including data correction, feature reduction, and image classification using machine learning techniques. She has coauthored the remote sensing textbook titled *Remote Sensing Digital Image Analysis* [Springer-Verlag, 3rd (1999) and 4th eds. (2006)]. Her research interests include remote sensing, hyperspectral image processing, and spatial data analysis.



Zheng Han received the B.Sc. degree in geomatics engineering from the China University of Petroleum (East China), Qingdao, China, in 2022, where he is currently pursuing the M.D. degree in surveying and mapping.

His research interests include urban impervious surface extraction and multisource remote sensing image fusion.

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